A Dialectica Model of the Lambek Calculus

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Originally published in the Proceedings of the Eighth Amsterdam Colloquium, 1991, pp. 445–461.
Retyped 2026.

This note discusses the Lambek Calculus (Lambek [1958], van Benthem [1988]) and some variants from the point-of-view of Linear Logic (Girard [1987]). The Lambek Calculus was first introduced, with the name of Syntactic Calculus, by J. Lambek in 1958 as an explanation of the mathematics of sentence structure. After a long period of ostracism, around 1980, the Syntactic Calculus, now called the Lambek Calculus, was taken up by logicians interested in Computational Linguistics, especially in the area of Categorial Grammar. The calculus was considered "like a logical system" but much too weak to be taken seriously as a logic. To quote from van Benthem [1988],

> . . . One fundamental system of this kind is the so-called 'Lambek Calculus', whose type-change rules show a close analogy with the inference rules of constructive propositional logic.

This situation could change drastically with the advent of Girard's Linear Logic (Girard [1987]). Girard showed that there is a full embedding, preserving proofs, of Intuitionistic Logic into Linear Logic with modality '!'. That means, in loose terms, that Linear Logic is a logic[1] as expressive as Intuitionistic Logic and the claim is that it is more precise, i.e. it makes finer distinctions. As the Lambek Calculus is the (multiplicative) fragment of non-commutative Intuitionistic Linear Logic, several interesting questions can be asked. One of them, posed by Morrill et al [1990] is whether we can extend the Lambek Calculus with a modality that does for the structural rule of (*exchange*) what the modality '!' does for the rules of (*weakening*) and (*contraction*). A very preliminary proposal, which answers this question affirmatively, is set forward in this paper. But it must be said from the start that the 'answer' is only provided in semantical terms. The proof theory of the systems considered should be investigated in future work. Another warning is that the perspective of this note is basically from Category Theory as a branch of Mathematics, so words like categories and functors are always meant in their mathematical, rather than linguistical or philosophical sense.

In the first section we recall Linear Logic and provide the transformations to show that the Lambek Calculus **L** really is the multiplicative fragment of (non-commutative) Intuitionistic Linear Logic. In the second section we describe the usual String Semantics for the Lambek Calculus **L** and generalise it, using a categorical perspective. In the third section we describe a Dialectica-style model for the Lambek Calculus. In the last section we discuss (semantically valid) modalities and hint at some future work.

I would like to thank Jan van Eijck for inviting me to give the talk that became this note, thereby gently 'forcing' me to think about the subject, as well as, for his generous hospitality.

[1] The whole issue of what a logic is *philosophically* is being eschewed here though.

I also would like to thank Martin Hyland, Harold Schellinx, Dirk Roorda, Mark Hepple, Glyn Morrill and Michael Moortgat for several useful discussions. Many of the ideas in this paper have been shaped by these discussions, but of course the mistakes are all mine. Finally I want to thank Jim Lambek for 'putting me right' about how completeness has nothing to do with the existence of two disjunctions, in the most friendly possible way.

1 From Linear Logic to the Lambek Calculus

Intuitionistic Linear Logic, henceforth **ILL**, was described in Girard and Lafont's [1987] TAPSOFT paper. One of the best ways of thinking about the system **ILL** is to start with Gentzen's Intuitionistic Calculus **LJ**, as for instance in Takeuti [1975] (page 19) and delete the structural rules of (*weakening*) and (*contraction*)

$$\frac{\Gamma \vdash A}{\Gamma, B \vdash A}\ (\textit{weakening}) \qquad \frac{\Gamma, B, B \vdash A}{\Gamma, B \vdash A}\ (\textit{contraction})$$

Without (*weakening*) and (*contraction*) the two possible formulations for the rule of conjunction,

$$\frac{\Gamma \vdash A \qquad \Gamma \vdash B}{\Gamma \vdash A \wedge B} \qquad \frac{\Gamma \vdash A \qquad \Delta \vdash B}{\Gamma, \Delta \vdash A \wedge B}$$

give rise to two different connectives, which Girard calls, respectively, the 'additive' and the 'multiplicative' conjunctions. Recall the rules and axioms for **ILL** in a sequent-calculus formulation.

Intuitionistic Linear Logic (ILL)

Axioms:

$$A \vdash A\ \ (\textit{identity}) \qquad \vdash I \qquad \Gamma \vdash 1 \qquad \Gamma, 0 \vdash A$$

Structural Rules:

$$\frac{\Gamma, A, B, \Gamma' \vdash C}{\Gamma, B, A, \Gamma' \vdash C}\ (\textit{permutation}) \qquad \frac{\Gamma \vdash A \qquad A, \Gamma' \vdash B}{\Gamma, \Gamma' \vdash B}\ (\textit{cut})$$

Logical Rules:
Multiplicatives:

$$(\textit{unit})\ \frac{\Gamma \vdash A}{\Gamma, I \vdash A}$$

$$(\otimes_l)\ \frac{\Gamma, A, B \vdash C}{\Gamma, A \otimes B \vdash C} \qquad (\otimes_r)\ \frac{\Gamma \vdash A \qquad \Gamma' \vdash B}{\Gamma, \Gamma' \vdash A \otimes B}$$

$$(\multimap_l)\ \frac{\Gamma \vdash A \qquad B, \Gamma' \vdash C}{\Gamma, A \multimap B, \Gamma' \vdash C} \qquad (\multimap_r)\ \frac{\Gamma, A \vdash B}{\Gamma \vdash A \multimap B}$$

Additives:

$$(\&_r)\ \frac{\Gamma \vdash A \qquad \Gamma \vdash B}{\Gamma \vdash A \,\&\, B} \qquad (\&_l)\ \frac{\Gamma, A \vdash C}{\Gamma, A \,\&\, B \vdash C} \qquad \frac{\Gamma, B \vdash C}{\Gamma, A \,\&\, B \vdash C}$$

$$(\oplus_l)\ \frac{\Gamma, A \vdash C \qquad \Gamma, B \vdash C}{\Gamma, A \oplus B \vdash C} \qquad (\oplus_r)\ \frac{\Gamma \vdash A}{\Gamma \vdash A \oplus B} \qquad \frac{\Gamma \vdash B}{\Gamma \vdash A \oplus B}$$

A difference between the sequent-calculus for traditional intuitionistic logic and the one above is that one thinks of Γ as a set in the traditional case, while here Γ means a multiset (or a bag) of assumptions.

Note that in the multiplicative fragment, we can put different contexts Γ and Γ' together – rules $(\otimes_r)$ and $(\multimap_l)$ – while in the additive fragment, we must have exactly the same context Γ – rules $(\&_r)$ and $(\oplus_r)$ – to apply a rule. Observe that, as the system **ILL** satisfies Cut-elimination, there is only one structural rule left, namely (*permutation*). Also in **ILL** we have only one (additive) disjunction '$\oplus$', which contrasts with Classical Linear Logic (Girard [1987]) or Full Intuitionistic Linear Logic (Hyland and de Paiva [1990]).

If we delete the rules for the additives (and their axioms) from the system above, we have the system **LP** studied by van Benthem [1988] and Moortgat [1990], that is the Lambek Calculus with Permutation. Note that this system could also be called $\mathbf{ILL}_m$, that is the multiplicative fragment of **ILL**. For the sake of completeness we list the axioms and rules of **LP**.

Lambek Calculus with Permutation (LP)

Axioms:

$$A \vdash A \quad (\textit{identity}) \qquad \vdash I$$

Structural Rules:

$$\frac{\Gamma, A, B, \Gamma' \vdash C}{\Gamma, B, A, \Gamma' \vdash C}\ (\textit{permutation}) \qquad \frac{\Gamma \vdash A \qquad A, \Gamma' \vdash B}{\Gamma, \Gamma' \vdash B}\ (\textit{cut})$$

Logical Rules:

$$(\textit{unit})\ \frac{\Gamma \vdash A}{\Gamma, I \vdash A}$$

$$(\otimes_l)\ \frac{\Gamma, A, B \vdash C}{\Gamma, A \otimes B \vdash C} \qquad (\otimes_r)\ \frac{\Gamma \vdash A \qquad \Gamma' \vdash B}{\Gamma, \Gamma' \vdash A \otimes B}$$

$$(\multimap_l)\ \frac{\Gamma \vdash A \qquad \Gamma', B \vdash C}{\Gamma, \Gamma', A \multimap B \vdash C} \qquad (\multimap_r)\ \frac{\Gamma, A \vdash B}{\Gamma \vdash A \multimap B}$$

Now if we remove the last structural rule of (*permutation*), which "means" the commutativity of $\otimes$, the rules for '$\multimap$' can be given in two (now different) guises, e.g.

$$(\rightharpoonup_r)\ \frac{\Gamma, A \vdash B}{\Gamma \vdash A \rightharpoonup B} \qquad (\leftharpoondown_r)\ \frac{A, \Gamma \vdash B}{\Gamma \vdash A \leftharpoondown B}$$

Thus we get two directional implications and the system is called the Lambek Calculus **L**. The rules and axioms of **L** are as follows:

Lambek Calculus (L)

Axioms:

$$A \vdash A \quad (identity)$$
$$\vdash I$$

Structural Rule:

$$\frac{\Gamma \vdash A \qquad \Gamma', A, \Delta \vdash B}{\Gamma', \Gamma, \Delta \vdash B}\ (cut)$$

Logical Rules:

$$(unit)\ \frac{\Gamma \vdash A}{\Gamma', I, \Gamma'' \vdash A}\ (\Gamma', \Gamma'' = \Gamma)$$

$$(\otimes_l)\ \frac{\Gamma, A, B, \Gamma' \vdash C}{\Gamma, A \otimes B, \Gamma' \vdash C} \qquad (\otimes_r)\ \frac{\Gamma \vdash A \qquad \Gamma' \vdash B}{\Gamma, \Gamma' \vdash A \otimes B}$$

$$(\rightharpoonup_l)\ \frac{\Gamma \vdash A \qquad \Gamma', B, \Delta \vdash C}{\Gamma', A \rightharpoonup B, \Gamma, \Delta \vdash C} \qquad (\rightharpoonup_r)\ \frac{\Gamma, A \vdash B}{\Gamma \vdash A \rightharpoonup B}$$

$$(\leftharpoondown_l)\ \frac{\Gamma \vdash A \qquad \Gamma', B, \Delta \vdash C}{\Gamma', \Gamma, A \leftharpoondown B, \Delta \vdash C} \qquad (\leftharpoondown_r)\ \frac{A, \Gamma \vdash B}{\Gamma \vdash A \leftharpoondown B}$$

Note that, due to the absence of permutation, Γ is not a multiset, but a sequence of assumptions. The lack of rule (*permutation*) splits the connective '$\multimap$' into two implications '$\rightharpoonup$' and '$\leftharpoondown$', which one can think of as *over* and *under* or predictive and retroactive implication. For that reason, we call '$\rightharpoonup$' *forward* and '$\leftharpoondown$' *back*. Lack of (*permutation*) also accounts for the small modifications in the rules (*cut*), (*unit*) and $(\otimes_l)$.

There are several variants of this calculus. The system above is called the *syntactic calculus* by Lambek in page 305 Lambek [1988] but in Lambek [1987] we have a version without the constant I. Moortgat [1990] presents a version without (explicit) tensors $\otimes$ (and I), which he calls the Lambek-Gentzen Calculus **L**. Moortgat also gives a Hilbert-style presentation of this logic, which he calls $\mathbb{Z}$ for Zielonka and says that the equivalence between $\mathbb{Z}$ and the Lambek-Gentzen system was shown by Buszkowski. Both Moortgat and Lambek consider briefly the system where the operation $\otimes$ – usually denoted by a dot '$\cdot$' – is *not* associative, but we will not do this here.

In Lambek [1987] we also found the definition,

Definition 1. *A* categorial grammar *of a language may be viewed as consisting of the syntactic calculus* **L** *freely generated from a finite set* $\{S, N, \dots\}$ *of basic types together with a dictionary which assigns to each word of the language a finite set of types composed from the basic types and* I *by the three binary operations,* $\otimes$, $\rightharpoonup$ *and* $\leftharpoondown$.

but we will only mention the definition of a categorial grammar and not try anything closer to real Linguistics in this note.

Moortgat and van Benthem also argue that for certain linguistics applications (non-peripheral extraction is one of the examples) one would like to have some amount of permutation in the system. We will try and say something about that in the last section. Before going on to semantics we present a last sequent-calculus system, very natural from the Linear Logic viewpoint, namely the Lambek calculus with additives, which we call **LA**.

Lambek Calculus with Additives (LA)

Axioms:

$$A \vdash A \quad (identity)$$
$$\vdash I$$
$$\Gamma \vdash 1 \qquad \Gamma, 0, \Gamma' \vdash A$$

Structural Rule:

$$\frac{\Gamma \vdash A \qquad \Gamma', A, \Delta \vdash B}{\Gamma', \Gamma, \Delta \vdash B}\ (cut)$$

Logical Rules:
Multiplicatives:

$$(unit)\ \frac{\Gamma, \Gamma' \vdash A}{\Gamma, I, \Gamma' \vdash A}$$

$$(\otimes_l)\ \frac{\Gamma, A, B, \Delta \vdash C}{\Gamma, A \otimes B, \Delta \vdash C} \qquad (\otimes_r)\ \frac{\Gamma \vdash A \qquad \Gamma' \vdash B}{\Gamma, \Gamma' \vdash A \otimes B}$$

$$(\rightharpoonup_l)\ \frac{\Gamma \vdash A \qquad \Gamma', B, \Delta \vdash C}{\Gamma', A \rightharpoonup B, \Gamma, \Delta \vdash C} \qquad (\rightharpoonup_r)\ \frac{\Gamma, A \vdash B}{\Gamma \vdash A \rightharpoonup B}$$

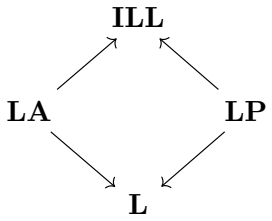
$$(\leftharpoondown_l)\ \frac{\Gamma \vdash A \qquad \Gamma', B, \Delta \vdash C}{\Gamma', \Gamma, A \leftharpoondown B, \Delta \vdash C} \qquad (\leftharpoondown_r)\ \frac{A, \Gamma \vdash B}{\Gamma \vdash A \leftharpoondown B}$$

Additives:

$$(\&_r)\ \frac{\Gamma \vdash A \qquad \Gamma \vdash B}{\Gamma \vdash A \,\&\, B} \qquad (\&_l)\ \frac{\Gamma, A, \Delta \vdash C}{\Gamma, A \,\&\, B, \Delta \vdash C} \qquad \frac{\Gamma, B, \Delta \vdash C}{\Gamma, A \,\&\, B, \Delta \vdash C}$$

$$(\oplus_l)\ \frac{\Gamma, A, \Delta \vdash C \qquad \Gamma, B, \Delta \vdash C}{\Gamma, A \oplus B, \Delta \vdash C} \qquad (\oplus_r)\ \frac{\Gamma \vdash A}{\Gamma \vdash A \oplus B} \qquad \frac{\Gamma \vdash B}{\Gamma \vdash A \oplus B}$$

Note that the system **LA** could also be called non-commutative intuitionistic linear logic. That is **ILL** = **LP** + **A**. As a summary of the systems discussed we draw a diagram



But remember that to obtain **LP** from **L** we add an inference rule, namely (*permutation*), whereas to obtain **LA** from **L** we add the additive connectives.

2 Semantics

The usual model for the logical system **L** (Lambek [1958]) is given by sets of expressions from a (generally taken as finite) non-empty universe V^+. This model is called the **String**

Semantics for the Lambek Calculus **L**. To describe it, given subsets A and B of V^+, define:

$$A \cdot B = \{xy \in V^+ \mid x \in A \text{ and } y \in B\}$$
$$C/B = \{x \in V^+ \mid \text{ for all } y \in B, xy \in C\}$$
$$A\backslash C = \{y \in V^+ \mid \text{ for all } x \in A, xy \in C\}$$

If you call $\mathcal{N}$ the set of all subsets of V^+, $\mathcal{N}$ has a natural order induced by *set containment*. Concatenation, represented by the dot in $A \cdot B$, gives $\mathcal{N}$ a (non-commutative) monoidal structure. Concatenation is associative, has identity given by the empty sequence, and is, in general, non-commutative. Given $A, B \in \mathcal{N}$

- $A \cdot (B \cdot C) = (A \cdot B) \cdot C$ (associativity of concatenation);
- If we denote by I the empty sequence $A \cdot I = I \cdot A = A$ (identity);
- $A \cdot B \not\leq B \cdot A$, in general.

Now, we want to generalise the structure above, taking our clues from Category Theory (MacLane [1971]). To do that we first recall the definition of an ordered (non-commutative) monoid.

Definition 2. *An* ordered (non-commutative) monoid *is a poset $(M, \leq)$ with a given compatible monoidal structure $(M, \circ, e)$. That is, a set M equipped with a binary relation '$\leq$' satisfying:*

- *$a \leq a$ for all a in M (*reflexivity*)*
- *$a \leq b$ and $b \leq c \Rightarrow a \leq c$ (*transitivity*)*
- *$a \leq b$ and $b \leq a \Rightarrow a = b$ (*antisymmetry*)*

together with a monoid structure $(\circ, e)$ consisting of a 'multiplication' $\circ\colon M \times M \to M$ and a distinguished object 'e' of M, such that the following hold:

- *$(a \circ b) \circ c = a \circ (b \circ c)$ (*associativity*)*
- *$a \circ e = e \circ a = a$ (*identity*)*

The structures are compatible in the sense that, if $a \leq b$, we have

- *$a \circ c \leq b \circ c$, for all c in M and $c \circ a \leq c \circ b$, for all c in M.*

We write an ordered (non-commutative) monoid as a quadruple $(M, \leq, \circ, e)$. Note that

$$(\mathcal{N}, \leq, \cdot, I)$$

in the string semantics above is an ordered (non-commutative) monoid as the order $\leq$ is compatible with the multiplication given by concatenation. But ordered monoids are not the structure we are after, we need some structure to model the Lambek "slashes" (or $\leftharpoonup$ and $\rightharpoonup$ in our notation). Following the tradition of lattice theory we define *left(right)-pseudocomplements* and more importantly the notion of a *biclosed poset*.

Definition 3. *Suppose M is an ordered monoid and $a, b \in M$. If there exists a largest $x \in M$ such that $a \circ x \leq b$ then this element is denoted $a \rightharpoonup b$ and it is called the* left pseudocomplement *of a* wrt *b. Similarly, if there exists a largest $x \in M$ such that $x \circ a \leq b$ then this element is denoted $a \leftharpoonup b$ and it is called the* right pseudocomplement *of a* wrt *b.*

A biclosed poset *is an ordered monoid $(M, \leq, \circ, e)$ such that $a \rightharpoonup b$ and $a \leftharpoonup b$ exist for all a and b in M. We write a biclosed poset as 6-tuple $(M, \leq, \circ, e, \rightharpoonup, \leftharpoonup)$.*

Note that in a biclosed poset $(M, \leq, \circ, e, \rightharpoonup, \leftharpoonup)$ the binary operations $\leftharpoonup$ and $\rightharpoonup$ are antimonotonic in their first coordinate, monotonic in their second coordinate and such that:

$$a \circ b \leq c \text{ iff } a \leq b \rightharpoonup c \text{ in } M \text{ and } b \circ a \leq c \text{ iff } a \leq c \leftharpoonup b \text{ in } M$$

The notion of a biclosed poset is the natural simplification to the poset set up of the notion of a (non-symmetric) monoidal biclosed category, see Eilenberg and Kelly [1965]. It is clear that $(\mathcal{N}, \leq, \cdot, I)$ with A/B and $A\backslash B$ defined for the string semantics is a biclosed poset. We now state some propositions, which are well-known facts:

FACT 1. *Any biclosed poset* $(M, \leq, \circ, e, \rightharpoonup, \leftharpoonup)$ *is a model for the Lambek Calculus* **L**.

By that we mean that if $\Gamma \vdash A$ is a derivable sequent in **L**, and $(\ )^0$ assigns to each basic type A an element A^0 of the biclosed poset M, then $\Gamma^0 \leq A^0$.

Note that if the monoidal operation '$\circ$' is commutative, a left pseudocomplement is also a right pseudocomplement, and the two implications '$\rightharpoonup$' and '$\leftharpoonup$' collapse back into a single one. The *commutative* structure corresponding to a biclosed poset is a *closed poset*, examples can be found in Lawvere [1973] and Flagg [1990]. Hence:

FACT 2. *Any (symmetric) closed poset* $(M, \leq, \circ, e, \multimap)$ *is a model for the Lambek Calculus with permutation* **LP**.

Recall that **LA** denotes the Lambek calculus with additives. The system **LA** satisfies cut-elimination, as the proof for **ILL** – with the modification made by Lambek to account for two implications – also work for **LA**. The calculus **LA** was given semantics by Morrill [1990], Moortgat [1990] and others. One way of adding additives to the string model above is defining:

$$A \cap B = \{x \in V^+ \mid x \in A \text{ and } x \in B\}$$
$$A \cup B = \{x \in V^+ \mid x \in A \text{ or } x \in B\}$$

To generalise the string semantics with additives, we could say:

FACT 3. *Any biclosed poset with (finite) meets and joins* $(M, \leq, \circ, e, \rightharpoonup, \leftharpoonup, \wedge, \vee)$ *is a model for the Lambek Calculus with additives* **LA**.

But note that adding booleans is stronger than adding additives, as for instance additives do not necessarily satisfy distributivity $A\&(B \oplus C) \ncong (A\&B) \oplus (A\&C)$ whereas booleans do (see Roorda [1991]). A biclosed poset with finite meets and joins is the natural simplification to the poset set up of the notion of a (non-symmetric) monoidal biclosed category with finite products and coproducts. For the sake of completeness we state:

FACT 4. *A (symmetric) closed poset with (finite) meets and joins* $(M, \leq, \circ, e, \multimap, \wedge, \vee)$ *is a model for the Lambek Calculus with additives and full permutation* **ILL**.

Note that a symmetric closed poset with meets and joins is almost a *quantale* as in Yetter [1990], i.e. it is a quantale without completeness. Many people have considered quantales as models for Intuitionistic Linear Logic, but the (algebraic) completeness of a quantale is not necessary to model the propositional logic and it distracts from the essential structure.

Two examples

In this short subsection we show that mathematical structures, which look superficially very different from the traditional models, can model **L** as well as strings and residuated semigroups. In the first example we calculate with matrices.

Matrices as a Model for L

Consider the set $\mathcal{N}$ of invertible matrices with entries from $\mathbb{N}$ the set of natural numbers,

$$\mathcal{N} = \{\text{invertible square matrices } k \times k \text{ with elements in } \mathbb{N}\}$$

We use Greek letters α, β for our matrices. Of course we have the usual operation of matrix multiplication, denoted

$$\otimes : \mathcal{N} \times \mathcal{N} \to \mathcal{N}$$

Recall that matrix multiplication is a non-commutative monoidal operation on $\mathcal{N}$. That is

- There is an identity for matrix multiplication I, simply the identity matrix of dimension k,
- For any matrix α, $\alpha \otimes I = I \otimes \alpha = \alpha$,
- Associativity of matrix multiplication $(\alpha \otimes \beta) \otimes \gamma = \alpha \otimes (\beta \otimes \gamma)$ holds and
- The matrix product is, in general, non-commutative $\alpha \otimes \beta \not\leq \beta \otimes \alpha$.

We can define a pre-order on $\mathcal{N}$, saying that

$$\alpha \leq \beta \qquad \text{iff} \qquad \alpha^{-1} \otimes \beta \leq I$$

Given any pre-order we have an induced partial order by taking equivalence classes, the usual construction. The order induced on $\mathcal{N}$ is compatible with the matrix multiplication. Thus $(\mathcal{N}, \leq, \otimes, I)$ is another non-commutative ordered monoid. Moreover, we can use the invertibility of our matrices to define two operations that make $\mathcal{N}$ a biclosed poset. We define two binary operations on elements of $\mathcal{N}$, $[(-) \rightharpoonup (-)]$ and $[(-) \leftharpoonup (-)]$, by saying that $\alpha \rightharpoonup \beta = \alpha^{-1} \otimes \beta$ and $\alpha \leftharpoonup \beta = \beta^{-1} \otimes \alpha$.

Note that these are operations of the form $\mathcal{N}^{op} \times \mathcal{N} \to \mathcal{N}$. By that we mean that the operations are antimonotonic in the first coordinate and monotonic in the second. Hence:

Proposition 1. *The ordered monoid of invertible $k \times k$ matrices over $\mathbb{N}$, $(\mathcal{N}, \leq, \otimes, I)$ with the operations of matrix multiplication '$\otimes$', and '$\leftharpoonup$' and '$\rightharpoonup$' defined above constitutes a biclosed poset.*

Thus $(\mathcal{N}, \leq, \otimes, I, \leftharpoonup, \rightharpoonup)$ is another model for the Lambek Calculus and there are several other models like modules of rings, etc. According to Lambek the idea for the Syntactic Calculus came from algebraic models. Actually it is easy to see that if one has a(n ordered) group, i.e. the possibility of inverting elements of an (non-symmetric) ordered monoid, we will have a biclosed poset.

$\mathbb{N}$ as a Model for LP

If we want to produce algebraic models for **LP**, the calculus with permutation, instead of the Lambek Calculus, we have many more 'familiar' structures, like, for instance, the set of natural numbers or the positive reals.

The set of the natural numbers $\mathbb{N}$ is clearly a (symmetric) *closed poset*, where conjunction '$\circ$' is given by addition of natural numbers, the identity for this conjunction is 0 and the internal-hom is given by truncated subtraction, as in Lawvere [1973], (also in de Paiva [1991a]). It is easy to prove the adjunction and another easy argument on Lindenbaum algebras (following Girard's proof) proves completeness of the models above[2].

[2] I recently became aware of Abrusci's ([1990]) paper where starting from the same perspective (Linear Logic) he explains the Lindenbaum algebra construction.

In the next section we describe another model for the Lambek Calculus, but this time a categorical one, that is, one where we take in consideration different proofs of the same entailment.

3 A Dialectica Construction

In this section we show that we can construct a Dialectica[3] category model of the Lambek Calculus in a straightforward way. The construction is meant both as evidence of the flexibility and powerfulness of the dialectica-like constructions and also as a way of, semantically, showing that we can define a modality that adds restricted permutation to the Lambek Calculus. The construction is given without all the details, hence it might look too complicated at first reading. The reader should be warned that some familiarity with Category Theory is assumed in this section.

Now suppose $\mathbb{C}$ is the category of sets and usual maps, so, in particular, $\mathbb{C}$ is a monoidal closed – actually cartesian closed – category with finite products and coproducts. Since $\mathbb{C}$ has finite products, it has a terminal object 1, any singleton set. It also has an initial object 0, the empty set. We write – as usual – $A \times B$ for the categorical product and $A + B$ for the coproduct in $\mathbb{C}$. Consider as well a biclosed poset $(\mathcal{N}, \leq, \otimes, e, \leftharpoonup, \rightharpoonup)$ in $\mathbb{C}$, you could think about it as the string semantics model above.

Note that the partial order on $\mathcal{N}$ induces an order on the hom-set $\mathbb{C}(X, \mathcal{N})$ given by

$$\text{for } f, g\colon X \to \mathcal{N} \quad f \leq g \qquad \text{iff} \qquad \forall x \in X \quad f(x) \leq g(x)$$

Using the data above we can define a (dialectica) category $\mathrm{Gir}_{\mathcal{N}}(\mathbb{C})$[4] as follows.

Definition 4. *Given a biclosed poset $(\mathcal{N}, \leq, \otimes, e, \leftharpoonup, \rightharpoonup)$ and $\mathbb{C}$ the category of* **Sets** *and usual functions, the category $\mathrm{Gir}_{\mathcal{N}}(\mathbb{C})$ consists of:*

- *Objects of $\mathrm{Gir}_{\mathcal{N}}(\mathbb{C})$ are triples (U, X, α) where α is a map in $\mathbb{C}$ of the form*

  $$U \times X \xrightarrow{\alpha} \mathcal{N}$$

  We call the object above A, because it is determined by the function α; and write it as $A = (U \overset{\alpha}{\leftrightarrow} X)$.

- *Morphisms in $\mathrm{Gir}_{\mathcal{N}}(\mathbb{C})$ from an object A to an object B of the form $V \times Y \overset{\beta}{\to} \mathcal{N}$ correspond to pairs of maps in $\mathbb{C}$, $f\colon U \to V$ and $F\colon Y \to X$ such that in the following diagram*

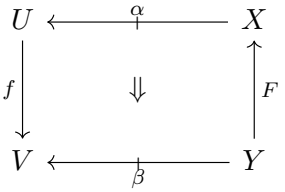
  $$\begin{array}{ccc} U \times Y & \xrightarrow{U \times F} & U \times X \\ {\scriptstyle f \times Y}\downarrow & & \downarrow{\scriptstyle \alpha} \\ V \times Y & \xrightarrow[\beta]{} & \mathcal{N} \end{array}$$

  we have $\alpha \circ (U \times F) \leq \beta \circ (f \times Y)$ as morphisms in $\mathbb{C}(U \times Y, \mathcal{N})$.

[3] Dialectica categories were introduced by the author in de Paiva [1987] as an internal categorical version of Gödel's Dialectica Interpretation. They turned out to be also an interesting family of models of Intuitionistic Linear Logic.

[4] We write $\mathrm{Gir}_{\mathcal{N}}(\mathbb{C})$ for this category. The author's thesis (de Paiva [1991b]) and de Paiva [1989] call the corresponding construction GC; Gir is the name we now use for it, as it is the Girard variant of the Dialectica construction.

We write morphisms in $\mathrm{Gir}_{\mathcal{N}}(\mathbb{C})$ as diagrams:



$$\forall u \in U, \forall y \in Y \quad \alpha(u, Fy) \leq \beta(fu, y)$$

Note that checking that the data above really defines a category is not totally trivial. The category $\mathrm{Gir}_{\mathcal{N}}(\mathbb{C})$ above is a variant of the construction in the author's thesis (de Paiva [1991b]) which is related to Chu's construction (Barr [1979]) and to $GAME_k$, (Lafont and Streicher [1991]).

Proposition 2. *The description above defines a category* $\mathrm{Gir}_{\mathcal{N}}(\mathbb{C})$.

Composition of morphisms (f, F) and (g, G) is given by the morphism (gf, FG). Identities are identities in each coordinate and associativity is given by the associativity of function composition in $\mathbb{C}$, that is **Sets**.

Having defined the category we want to use as a model for the Lambek Calculus, we now embark in a series of boring and relatively easy calculations to show that this category is monoidal biclosed. We start by defining a monoidal structure on $\mathrm{Gir}_{\mathcal{N}}(\mathbb{C})$.

Definition 5. *Given two objects* $(U \overset{\alpha}{\leftrightarrow} X)$ *and* $(V \overset{\beta}{\leftrightarrow} Y)$ *in* $\mathrm{Gir}_{\mathcal{N}}(\mathbb{C})$ *we define* $A \otimes B$ *their tensor product as follows:*

$$A \otimes B = (U \times V \overset{\alpha \otimes \beta}{\longleftarrow\!\!\!\!+\!\!\!\!-} X^V \times Y^U)$$

The morphism "$\alpha \otimes \beta$*" is given intuitively by* $\alpha \otimes \beta(u, v, f, g) = \alpha(u, fv) \otimes \beta(v, gu)$*, where the second* $\otimes$ *refers to the (non-commutative) monoidal operation in* $\mathcal{N}$*.*

To give a formal definition of $\alpha \otimes \beta$ consider the map

$$U \times V \times X^V \times Y^U \xrightarrow{\text{“}\langle ev, ev\rangle\text{”}} U \times Y \times V \times X \xrightarrow{\alpha \times \beta} \mathcal{N} \times \mathcal{N} \xrightarrow{\otimes} \mathcal{N}$$

Note that we are using the cartesian closed – hence commutative – structure of **Sets** – to write X^V and Y^U – and the monoidal structure of $\mathcal{N}$ to define $\alpha \otimes \beta$.

This operation clearly defines a bifunctor, which is a tensor product. Associativity is straightforward and the object $I = (1 \overset{I}{\leftrightarrow} 1)$ – the map $1 \times 1 \overset{I}{\to} \mathcal{N}$, which only picks the identity I in $\mathcal{N}$ – is the identity for this tensor product.

Note that the 'carrier' of the tensor product is the same for $A \otimes B$ and $B \otimes A$, due to commutativity of products in $\mathbb{C}$, but as objects in $\mathrm{Gir}_{\mathcal{N}}(\mathbb{C})$ they are different, so the tensor product in $\mathrm{Gir}_{\mathcal{N}}(\mathbb{C})$ is *not* commutative.

One observation is that if one tries to get a (Dialectica) model for the Lambek calculus using a (non-symmetric) monoidal closed category $\mathbb{C}$ that leads into 'braiding' problems studied by Joyal and Street [1986], which seems worth investigating.

Actually this odd-looking tensor product, analogous to the one in $\mathrm{Gir}(\mathbb{C})$ cf. de Paiva [1989], is the right one to prove monoidal-biclosedness of $\mathrm{Gir}_{\mathcal{N}}(\mathbb{C})$ with respect to the two internal-homs. To define the internal-homs:

Definition 6. *Given two objects* $(U \overset{\alpha}{\leftrightarrow} X)$ *and* $(V \overset{\beta}{\leftrightarrow} Y)$ *in* $\mathrm{Gir}_{\mathcal{N}}(\mathbb{C})$ *we define* $[A, B]_l$ *their left – respectively right – internal hom as follows:*

$$[A, B]_l = (V^U \times X^Y \overset{\alpha \multimap \beta}{\longleftarrow\!\!\!\!+\!\!\!\!-} U \times Y)$$

$$[A, B]_r = (V^U \times X^Y \overset{\alpha \multimap \beta}{\longleftarrow\!\!\!\!+\!\!\!\!-} U \times Y)$$

The morphisms "$\alpha \rightharpoonup \beta$" and "$\alpha \leftharpoonup \beta$" are given intuitively by $(\alpha \rightharpoonup \beta)(f, F, u, y) = \alpha(u, Fy) \rightharpoonup \beta(fu, y)$, respectively, $(\alpha \leftharpoonup \beta)(f, F, u, y) = \alpha(u, Fy) \leftharpoonup \beta(fu, y)$, where '$\rightharpoonup$' and '$\leftharpoonup$' are the corresponding 'internal-homs' or 'slashes' in $\mathcal{N}$.

Having defined internal homs and a tensor product we state and prove the desired theorem.

Theorem 1. *The category* $\mathrm{Gir}_{\mathcal{N}}(\mathbb{C})$ *is a (non-symmetric) monoidal biclosed category.*

Proof. The proof is very simple, one has to verify the natural isomorphisms

$$Hom_{\mathrm{Gir}_{\mathcal{N}}(\mathbb{C})}(A \otimes B, C) \cong Hom_{\mathrm{Gir}_{\mathcal{N}}(\mathbb{C})}(A, [B, C]_l)$$
$$Hom_{\mathrm{Gir}_{\mathcal{N}}(\mathbb{C})}(B \otimes A, C) \cong Hom_{\mathrm{Gir}_{\mathcal{N}}(\mathbb{C})}(A, [C, B]_r)$$

This can be done by looking at the diagrams

$$\begin{array}{ccc} U \times V & \xleftarrow{\alpha\otimes\beta} & X^V \times Y^U \\ {\scriptstyle f}\downarrow & & \uparrow{\scriptstyle \langle F_1, F_2\rangle} \\ W & \xleftarrow[\gamma]{} & Z \end{array} \qquad \begin{array}{ccc} U & \xleftarrow{\alpha} & X \\ {\scriptstyle \langle f, F_2\rangle}\downarrow & & \uparrow{\scriptstyle F_1} \\ W^V \times Y^Z & \xleftarrow[\beta \rightharpoonup \gamma]{} & V \times Z \end{array}$$

and

$$\begin{array}{ccc} V \times U & \xleftarrow{\beta\otimes\alpha} & Y^U \times X^V \\ {\scriptstyle f}\downarrow & & \uparrow{\scriptstyle \langle F_1, F_2\rangle} \\ W & \xleftarrow[\gamma]{} & Z \end{array} \qquad \begin{array}{ccc} U & \xleftarrow{\alpha} & X \\ {\scriptstyle \langle f, F_2\rangle}\downarrow & & \uparrow{\scriptstyle F_1} \\ W^V \times Y^Z & \xleftarrow[\gamma \leftharpoonup \beta]{} & V \times Z \end{array}$$

and calculating (the hard bit) the sums.

If the morphism $(f, \langle F_1, F_2\rangle)$ is in $Hom(A \otimes B, C)$, then we know $(\alpha \otimes \beta) \leq \gamma$, that is

$$(\alpha \otimes \beta)(u, v, F_1 z, F_2 zu) \leq \gamma(f(u, v), z)$$

This means that $\alpha(u, F_1 zv) \otimes \beta(v, F_2 zu) \leq \gamma(f(u, v), z)$ in $\mathcal{N}$. But $\mathcal{N}$ is a biclosed poset, which implies that $\alpha \leq \beta \rightharpoonup \gamma$ or $\alpha(u, F_1 zv) \leq \beta(v, F_2 zu) \rightharpoonup \gamma(f(u, v), z)$. To show that the corresponding morphism $(\langle f, \overline{F_2}\rangle, \overline{F_1})$ is in $Hom(A, [B, C]_l)$ we have to show $\alpha(u, F_1(v, z)) \leq (\beta \rightharpoonup \gamma)(\langle f, F_2\rangle u, \langle v, z\rangle)$ that means $\alpha(u, F_1(v, z)) \leq (\beta \rightharpoonup \gamma)(fu, F_2 u, v, z)$. As $(\beta \rightharpoonup \gamma)(fu, F_2 u, v, z) = \beta(v, F_2 uz) \rightharpoonup \gamma(fuv, z)$ we know what we need, if transposing is allowed in the category $\mathbb{C}$. Similarly for the other isomorphism. □

That is all we need to show that the category $\mathrm{Gir}_{\mathcal{N}}(\mathbb{C})$ with the structure described is a model of the Lambek Calculus.

FACT 5. *A monoidal biclosed category* $\mathbb{M}$ *is a model for the Lambek Calculus* **L**.

The notion of a monoidal biclosed category reduces to the notion of a biclosed poset in the category of partial orders. Thus fact 5 is a generalisation of fact 1.

FACT 6. *The category* $\mathrm{Gir}_{\mathcal{N}}(\mathbb{C})$ *is a sound model for the Lambek Calculus* **L**, *meaning that if a sequent* $\Gamma \vdash A$ *is proved in* **L**, *there is a morphism* $f\colon |\Gamma| \to |A|$ *in the category.*

Atomic formulae are objects of $\mathrm{Gir}_{\mathcal{N}}(\mathbb{C})$, $\otimes$ is the non-commutative conjunction and '$\leftharpoonup$' and '$\rightharpoonup$' are the left and right internal-homs. The existence of the adjunction shows the validity of the inference rules for $\leftharpoonup$ and $\rightharpoonup$. □

The model $\mathrm{Gir}_{\mathcal{N}}(\mathbb{C})$ above is an easy model to calculate with as it is made out of sets, but it is not the syntactical traditional one. Recall that as we mentioned before there are

lots of other algebraic models. But the reason to give the construction above – admittedly not the easiest algebraic model – is that we can define a non-trivial of course '!' operator in $\mathrm{Gir}_{\mathcal{N}}(\mathbb{C})$, more about that in the next section.

Moreover we can define additives $A \,\&\, B$ and $A \oplus B$ very easily and independently of the $\mathcal{N}$ chosen.

Proposition 3. *The category* $\mathrm{Gir}_{\mathcal{N}}(\mathbb{C})$ *has binary (hence finite) products and coproducts given by*

$$A \,\&\, B = (U \times V \xleftarrow{\alpha \& \beta} X + Y)$$

$$A \oplus B = (U + V \xleftarrow{\alpha \oplus \beta} X \times Y).$$

Note that $\alpha \,\&\, \beta$ and $\alpha \oplus \beta$ actually only use either 'α' or 'β', as the coproduct forces us to choose, say between x and y, in the product case.

FACT 7. *The category* $\mathrm{Gir}_{\mathcal{N}}(\mathbb{C})$ *is a model for the Lambek Calculus with additives* **LA**.

The attentive reader will have noticed that we could state three other (well-known) facts, to the effect that symmetric monoidal closed categories are categorical models for **LP**, monoidal biclosed categories with (finite) products and coproducts are categorical models for **LA** and symmetric monoidal closed categories with products and coproducts (*linear categories* of Girard and Lafont [1987]) are categorical models for **ILL**.

4 Modalities

We start our discussion of modalities by recalling the rules for the modality "!" in usual *commutative* Intuitionistic Linear Logic. These are:

$$\frac{\Gamma, A \vdash B}{\Gamma, !A \vdash B}\ (\textit{dereliction}) \qquad \frac{\Gamma \vdash B}{\Gamma, !A \vdash B}\ (\textit{weakening})$$

$$\frac{\Gamma, !A, !A \vdash B}{\Gamma, !A \vdash B}\ (\textit{contraction}) \qquad \frac{!\Gamma \vdash A}{!\Gamma \vdash !A}\ (!)$$

As observed by several people, the four rules above fall neatly into two pairs. One pair, namely (*weakening*) and (*contraction*), has to do with putting back into the logic, in a controlled way, contraction and weakening for specific formulae and the other pair – (*dereliction*) and (!) – make '!' look like the (necessitation) $\Box$ modal operator of **S4**.

In categorical models one usually interprets '!' as a functor with a natural comonad structure, while the objects $!A$ have a natural comonoid structure with respect to the tensor product of the category. The comonad structure of '!' is used to interpret the S4-like rules for the modality !, but this interpretation also seems to use the fact that there are additives in the logic. Thus if one knows that

$$!(A \,\&\, B) \cong !A \otimes !B \text{ and } !0 \cong 1$$

and that there exists a comonad '!', we can reason as follows. There is always a map $!A \xrightarrow{\delta} !!A$ and if $!G \xrightarrow{f} A$, then we can apply the functor '!' to f, to get $!!G \xrightarrow{!f} !A$ and precompose it with $!G \xrightarrow{\delta} !!G$ to get $!G \longrightarrow !A$, which shows the rule (!) is satisfied. For this we are assuming that $!\Gamma \vdash A$ corresponds to a morphism $!G \to A$, and that

$$!\Gamma = !G_1 \otimes !G_2 \otimes \cdots \otimes !G_k \cong !(G_1 \,\&\, G_2 \,\&\, \ldots G_k).$$

Thus we are using that 'commas' really are instances of the connective '$\otimes$' and that the sequence $!G_1, !G_2, \ldots, !G_n$ can be transformed into '$!$' applied to the object $G_1 \& G_2 \cdots \& G_n$.

The reasoning above seems to hint at the usefulness of treating additives as well as multiplicatives, if one wants to re-introduce the structural rules into the system. It may not be necessary, but it certainly makes it semantically easier, as we shall see.

Some of the work done in non-commutative Linear Logic, Yetter [1990], Lincoln et al [1990] has kept a (weaker) notion of commutativity given by circular permutations. For the linguistic purposes envisaged for the Lambek calculus, the idea of allowing circular permutations is very odd. But for linguistical purposes no one is interested in having a single dualising element, admittedly at least in Yetter's paper, the main reason for the circular permutations rule.

It seems clear that formulas of the type $!A$, that we are allowed to contract and weaken over, should commute with any other formula, not only with some special ones. Also additives commute.

Hence what follows is a preliminary proposal, based mainly on Barry and Morrill [1990] and Yetter's [1990] ideas. We start with the calculus **LA** the Lambek calculus with additives, we add a modality like Yetter's 'κ' – *kappa* – with its rules and call the new calculus $\mathbf{LA}_\kappa$, for Lambek calculus with additives and (very restricted) permutation κ. Then we add Girard and Lafont's [1987] four rules for the modality '$!$' and call the resulting system $\mathbf{LA}_{\kappa!}$.

The reason for doing that is merely semantical, as we can interpret these modalities κ and $!$ in a variant of the Dialectica category $\mathrm{Gir}_{\mathcal{N}}(\mathbb{C})$ described in the previous section.

Lambek Calculus with Additives and κ ($\mathbf{LA}_\kappa$)

Axioms:

$$A \vdash A \quad (\textit{identity})$$

$$\vdash I$$

$$\Gamma \vdash 1 \qquad \Gamma, 0, \Gamma' \vdash A$$

Structural Rule:

$$\frac{\Gamma \vdash A \qquad \Gamma', A, \Delta \vdash B}{\Gamma', \Gamma, \Delta \vdash B}\ (\textit{cut})$$

Logical Rules:
Multiplicatives:

$$(\textit{unit})\ \frac{\Gamma \vdash A}{\Gamma', I, \Gamma'' \vdash A}$$

$$(\otimes)\ \frac{\Gamma, A, B, \Delta \vdash C}{\Gamma, A \otimes B, \Delta \vdash C} \qquad (\otimes)\ \frac{\Gamma \vdash A \qquad \Gamma' \vdash B}{\Gamma, \Gamma' \vdash A \otimes B}$$

$$(\multimap_l)\ \frac{\Gamma \vdash A \qquad \Gamma', B, \Delta \vdash C}{\Gamma', A \multimap B, \Gamma, \Delta \vdash C} \qquad (\multimap_r)\ \frac{\Gamma, A \vdash B}{\Gamma \vdash A \multimap B}$$

$$(\multimap_l)\ \frac{\Gamma' \vdash A \qquad \Gamma, B, \Delta \vdash C}{\Gamma, \Gamma', A \multimap B, \Delta \vdash C} \qquad (\multimap_r)\ \frac{A, \Gamma \vdash B}{\Gamma \vdash A \multimap B}$$

Additives:

$$(\&_r)\ \frac{\Gamma \vdash A \qquad \Gamma \vdash B}{\Gamma \vdash A \,\&\, B} \qquad (\&_l)\ \frac{\Gamma, A, \Delta \vdash C}{\Gamma, A \,\&\, B, \Delta \vdash C} \qquad \frac{\Gamma, B, \Delta \vdash C}{\Gamma, A \,\&\, B, \Delta \vdash C}$$

$$(\oplus_l)\ \frac{\Gamma, A, \Delta \vdash C \qquad \Gamma, B, \Delta \vdash C}{\Gamma, A \oplus B, \Delta \vdash C} \qquad (\oplus_r)\ \frac{\Gamma \vdash A}{\Gamma \vdash A \oplus B} \qquad \frac{\Gamma \vdash B}{\Gamma \vdash A \oplus B}$$

Rules for κ (kappa)

$$\frac{\Gamma, A, \kappa B, \Gamma' \vdash C}{\Gamma, \kappa B, A, \Gamma' \vdash C}\ (e1) \qquad \frac{\Gamma, \kappa A, B, \Gamma' \vdash C}{\Gamma, B, \kappa A, \Gamma' \vdash C}\ (e2)$$

$$\frac{\Gamma, A, \Delta \vdash C}{\Gamma, \kappa A, \Delta \vdash C}\ (der_k) \qquad \frac{\kappa\Gamma \vdash C}{\kappa\Gamma \vdash \kappa C}\ (\kappa)$$

Note that the difference between the systems ($\mathbf{LA}_\kappa$) and **ILL** is that permutation is restricted in ($\mathbf{LA}_\kappa$) to the formulas marked with a kappa. Also note that the rules for kappa are like S4-rules.

Recall that $\mathrm{Gir}_{\mathcal{N}}(\mathbb{C})$ models **LA** for any biclosed poset $\mathcal{N} = (N, \circ, \leq, e, \rightharpoonup, \leftharpoonup)$. In particular, we can take as $\mathcal{N}$ a (non-commutative) quantale Q. That is not really necessary, but simplifies parts of the exposition, as we can quote from Yetter's paper. Recall that by proposition 1.9 in page 44 of Yetter [1990], if Q is a quantale, there is a unique maximal central modality in Q, denoted $c_m \colon Q \to Q$. Use this modality c_m to define a modality κ in $\mathrm{Gir}_{\mathcal{N}}(\mathbb{C})$ as follows:

Definition 7. *Given a category* $\mathrm{Gir}_{\mathcal{N}}(\mathbb{C})$*, where* $\mathcal{N}$ *is a quantale* Q *we define*

$$\kappa \colon \mathrm{Gir}_{\mathcal{N}}(\mathbb{C}) \to \mathrm{Gir}_{\mathcal{N}}(\mathbb{C})$$

which acts on carriers of objects of $\mathrm{Gir}_{\mathcal{N}}(\mathbb{C})$ *as the identity endofunctor. Thus*

$$\kappa(U \overset{\alpha}{\leftrightarrow} X) = (U \overset{c_m\alpha}{\leftrightarrow} X)$$

where $U \times X \xrightarrow{\alpha} Q$*, a morphism in* $\mathbb{C}$*, is taken by* κ *to the composition*

$$U \times X \xrightarrow{\alpha} Q \xrightarrow{c_m} Q$$

and c_m *is the corresponding* unique maximal central *modality in* Q*.*

Moreover it is very easy to see that:

Proposition 4. *The endofunctor* $\kappa \colon \mathrm{Gir}_{\mathcal{N}}(\mathbb{C}) \to \mathrm{Gir}_{\mathcal{N}}(\mathbb{C})$ *has a natural structure of comonad.*

The natural transformations are induced by the 'natural transformations' in Q. All of this can be made much more concrete if one takes a quantale of relations (cf. Brown and Gurr [1991] for instance). The point here is that, the same way the completeness of a quantale is *not* necessary to model propositional Linear Logic, the completeness of a quantale is not necessary to model the existence of an of course '!' operator. But completeness is an added bonus (extra structure), which makes some calculations simpler. Because the modality is central it makes the kappa rules sound, cf. Yetter. Thus we have a proposition:

Proposition 5. *The category* $\mathrm{Gir}_{\mathcal{N}}(\mathbb{C})$ *– where* $\mathcal{N}$ *is a quantale* Q *– together with the comonad* κ *defined above is a categorical model for the Lambek Calculus with additives and restricted permutation or system* $\mathbf{LA}_\kappa$*.*

Now we add to the system $\mathbf{LA}_\kappa$ the Girard's rules for the modality '!'.

Lambek Calculus with Additives, κ and '!' ($\mathbf{LA}_{\kappa!}$)

Axioms:

$$A \vdash A \quad (identity)$$

$$\vdash I$$

$$\Gamma \vdash 1 \qquad \Gamma, 0, \Gamma' \vdash A$$

Structural Rule:

$$\frac{\Gamma \vdash A \qquad \Gamma', A, \Delta \vdash B}{\Gamma', \Gamma, \Delta \vdash B}\ (cut)$$

Logical Rules:
Multiplicatives:

$$(unit)\ \frac{\Gamma \vdash A}{\Gamma', I, \Gamma'' \vdash A}$$

$$(\otimes)\ \frac{\Gamma, A, B, \Delta \vdash C}{\Gamma, A \otimes B, \Delta \vdash C} \qquad (\otimes)\ \frac{\Gamma \vdash A \qquad \Gamma' \vdash B}{\Gamma, \Gamma' \vdash A \otimes B}$$

$$(\rightharpoonup_l)\ \frac{\Gamma \vdash A \qquad \Gamma', B, \Delta \vdash C}{\Gamma', A \rightharpoonup B, \Gamma, \Delta \vdash C} \qquad (\rightharpoonup_r)\ \frac{\Gamma, A \vdash B}{\Gamma \vdash A \rightharpoonup B}$$

$$(\leftharpoondown_l)\ \frac{\Gamma \vdash A \qquad \Gamma', B, \Delta \vdash C}{\Gamma', \Gamma, A \leftharpoondown B, \Delta \vdash C} \qquad (\leftharpoondown_r)\ \frac{A, \Gamma \vdash B}{\Gamma \vdash A \leftharpoondown B}$$

Additives:

$$(\&_r)\ \frac{\Gamma \vdash A \qquad \Gamma \vdash B}{\Gamma \vdash A \,\&\, B} \qquad (\&_l)\ \frac{\Gamma, A, \Delta \vdash C}{\Gamma, A \,\&\, B, \Delta \vdash C} \qquad \frac{\Gamma, B, \Delta \vdash C}{\Gamma, A \,\&\, B, \Delta \vdash C}$$

$$(\oplus_l)\ \frac{\Gamma, A, \Delta \vdash C \qquad \Gamma, B, \Delta \vdash C}{\Gamma, A \oplus B, \Delta \vdash C} \qquad (\oplus_r)\ \frac{\Gamma \vdash A}{\Gamma \vdash A \oplus B} \qquad \frac{\Gamma \vdash B}{\Gamma \vdash A \oplus B}$$

Rules for κ (kappa)

$$\frac{\Gamma, A, \kappa B, \Gamma' \vdash C}{\Gamma, \kappa B, A, \Gamma' \vdash C}\ (e1) \qquad \frac{\Gamma, \kappa A, B, \Gamma' \vdash C}{\Gamma, B, \kappa A, \Gamma' \vdash C}\ (e2)$$

$$\frac{\Gamma, A, \Delta \vdash C}{\Gamma, \kappa A, \Delta \vdash C}\ (der_k) \qquad \frac{\kappa\Gamma \vdash C}{\kappa\Gamma \vdash \kappa C}\ (\kappa)$$

Rules for ! (of course!)

$$\frac{\Gamma, A \vdash B}{\Gamma, !A \vdash B}\ (dereliction) \qquad \frac{\Gamma \vdash B}{\Gamma, !A \vdash B}\ (weakening)$$

$$\frac{\Gamma, !A, !A \vdash B}{\Gamma, !A \vdash B}\ (contraction) \qquad \frac{!\Gamma \vdash A}{!\Gamma \vdash !A}\ (!)$$

Rules connecting κ and !\[5\]

$$\frac{\Gamma, !A, \Gamma' \vdash B}{\Gamma, \kappa A, \Gamma' \vdash B}\ (!\kappa_l) \qquad\qquad \frac{\Gamma \vdash \kappa C}{\Gamma \vdash !C}\ (!\kappa_r)$$

The next step is to define a modality

$$!\colon \mathrm{Gir}_{\mathcal{N}}(\mathbb{C}) \to \mathrm{Gir}_{\mathcal{N}}(\mathbb{C})$$

which given an object $A = (U \overset{\alpha}{\longleftrightarrow} X)$ returns an object $(U \overset{!\alpha}{\longleftrightarrow} X^{*U})$. This can be shown to be an endofunctor with a natural comonad structure inherited from the monoidal structure of X^{*U}, cf. de Paiva [1991b]. There are lots of details to check, but as the category $\mathbb{C}$ is **Sets**, which is cartesian closed, the constructions above work and give you a comonad and comonoids in $\mathrm{Gir}_{\mathcal{N}}(\mathbb{C})$.

After checking all the details, we should have:

Proposition 6. *The category* $\mathrm{Gir}_{\mathcal{N}}(\mathbb{C})$ *is a categorical model for the Lambek Calculus with additives, restricted permutation* κ *and Girard modality '!' (of course!).*

As we mentioned before, this is all done semantically, it would be very interesting to check what happens with the Proof Theory of the systems ($\mathbf{LA}_\kappa$) and ($\mathbf{LA}_{\kappa!}$). On the one hand, the proof theory should not be too complicated, as we do not have the *par* operator interacting with linear implication, but on the other hand, people interested in proof-nets find weak forms of commutativity like (*cycling*) extremely natural.

Conclusions

If one takes the linguistics motivation seriously, the modality κ above does not seem exactly what is wanted. There might be some use in having an operator κ, but perhaps it would be more useful to have a modality which did a *very* limited amount of permutation. In particular it would be interesting to produce rules for a modality that applied to two formulas A and B said something like *It is known that* A *and* B *commute*. The modality κ above says something like κA *commutes with everything*.

There were some other attempts at less constraining permutative modalities and the reader should consult Barry et al for several possibilities. But the interesting aspect of the modality κ is that we can model it without too much difficulty. Also interesting is that it makes a sort of "bridge" between the commutative world and the non-commutative one. This bridge is of the same type as the bridge one has ("of course!") between resource-aware and traditional systems. Maybe one should investigate κ's proof theory, but our main concern at the moment is with *term calculi*, similar to the λ-calculus, but more accurate, that would model "exactly" other logical systems in the spirit of Linear Logic.

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